

A high order Discontinuous Galerkin – Fourier incompressible 3D Navier–Stokes solver with rotating sliding meshes

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ABSTRACT

We present the development of a sliding mesh capability for an unsteady high order ($order \geq 3$) h/p Discontinuous Galerkin solver for the three-dimensional incompressible Navier–Stokes equations. A high order sliding mesh method is developed and implemented for flow simulation with relative rotational motion of an inner mesh with respect to an outer static mesh, through the use of curved boundary elements and mixed triangular–quadrilateral meshes.

A second order stiffly stable method is used to discretise in time the Arbitrary Lagrangian–Eulerian form of the incompressible Navier–Stokes equations. Spatial discretisation is provided by the Symmetric Interior Penalty Galerkin formulation with modal basis functions in the x – y plane, allowing hanging nodes and sliding meshes without the requirement to use *mortar* type techniques. Spatial discretisation in the z -direction is provided by a purely spectral method that uses Fourier series and allows computation of spanwise periodic three-dimensional flows. The developed solver is shown to provide high order solutions, second order in time convergence rates and spectral convergence when solving the incompressible Navier–Stokes equations on meshes where fixed and rotating elements coexist.

In addition, an exact implementation of the no-slip boundary condition is included for curved edges; circular arcs and NACA 4-digit airfoils, where analytic expressions for the geometry are used to compute the required metrics.

The solver capabilities are tested for a number of two dimensional problems governed by the incompressible Navier–Stokes equations on static and rotating meshes: the Taylor vortex problem, a static and rotating symmetric NACA0015 airfoil and flows through three bladed cross-flow turbines. In addition, three dimensional flow solutions are demonstrated for a three bladed cross-flow turbine and a circular cylinder shadowed by a pitching NACA0012 airfoil.

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1. Introduction

Problems where the forces on rotating/oscillating geometries are to be predicted are common in engineering and fluid–structure interaction problems. Examples are flows around isolated rotating bodies and airfoils, turbomachinery applications, insect flight aerodynamics, unmanned air vehicles and more recently flows around renewable energy devices; wind and tidal turbines. These flows are characterised by long wake structures, vortex shedding and stalled flows associated with flow unsteadiness. To accurately simulate wakes and vortex structures and their evolution, high order (typically ≥ 3)

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numerical methods (i.e. h/p conformal spectral and h/p non-conformal Discontinuous Galerkin methods) are preferred since dissipation and dispersion errors are minimum [1] when compared to low order (typically ≤ 3) methods. Furthermore, for smooth problems, the exponential decay of the error with polynomial enrichment (p -refinement) in high order methods as opposed to the fixed decay rate characteristic of low order methods (i.e. h -refinement only) renders high order methods particularly attractive to obtain accurate solutions for flows where viscosity limits discontinuities (i.e. elliptic type equations) as in the incompressible Navier–Stokes (NS) equations.

For conformal discretisations (i.e. classic low order finite elements or h/p spectral type methods), if geometric discontinuities arise (e.g. hanging nodes), a *geometric incompatibility* [1] is created since C^0 continuity cannot be ensured across elements. Similarly, when neighbouring elements have different numbers of degrees of freedom (e.g. mixed triangular–quadrilateral mesh), a *functional incompatibility* [1] is created. Over the past few decades methods to overcome these difficulties have been developed in the framework of conformal discretisations, examples are *iterative patching*, *constrained approximation* or *mortar patching* (see [1] for a review). A natural alternative to overcome these incompatibilities is to relax the continuity condition across elements, which leads to the Discontinuous Galerkin (DG) finite element approach.

The DG finite element method can be seen as an extension of h/p spectral methods where the C^0 continuity requirement across element boundaries is relaxed or also as a high order finite volume method with compact stencil. Contrary to conformal finite elements or h/p spectral methods, DG methods are locally conservative by construction. As in spectral methods, high-order polynomials can be used within each element allowing exponential convergence, whilst discontinuities in the solution are permitted at element interfaces.

It has been argued that DG methods are prohibitively expensive, when compared to conformal spatial discretisations (e.g. h/p spectral methods), since additional degrees of freedom (DOF) arise from the discontinuities between elements. However, the relative number of the additional boundary degrees of freedom to internal degrees of freedom for each element, decreases rapidly for high polynomials. To exemplify this, let us consider a two dimensional problem and a mesh with N^{el} triangular elements with polynomial spaces of order k . On the one hand, if a h/p spectral discretisation is considered, the number of global degrees of freedom required is $DOF_{spectral} \approx \frac{N^{el}}{2} k^2$ (see [2]). On the other hand, a DG discretisation leads to a global number of DOF: $DOF_{DG} \approx \frac{N^{el}}{2} (k+1)(k+2) = \frac{N^{el}}{2} k^2 \left(1 + \frac{3}{k} + \frac{2}{k^2}\right)$ (see Section 2.2.6 for details). The last expression shows that for high polynomial orders k , the number of degrees of freedom for the DG discretisation tends to $DOF_{DG} \approx \frac{N^{el}}{2} k^2$. This simple analysis shows that for high polynomial orders $DOF_{DG} \approx DOF_{spectral}$ and the additional cost of DG methods is not overwhelming.

DG formulations to solve the incompressible NS equations have seen increased popularity over recent years as evidenced by the number of publications on the topic, (e.g. [3–9] and the authors' [10]). In [10], the authors presented the development of a DG code that uses the Symmetric Interior Penalty Galerkin (SIPG) formulation for solving the unsteady 2D NS equations using straight sided triangular elements. Simulation results from the solver were shown to be in good agreement with experimental results and results from a h/p conformal spectral code. The present implementation is an extension of the previous work to enable solutions on hybrid meshes (i.e. mixed triangular and quadrilateral elements) with rotationally sliding meshes and with curved boundary conditions. In addition, Fourier series applied orthogonally to the 2D DG plane are used to enable 3D flow solutions for problems with spanwise geometric homogeneity. These developments have been implemented and tested and results are reported herein.

The sliding mesh technique allows for mesh motion where an inner mesh zone region rotates with respect to an outer static mesh. This relative motion creates hanging nodes at the interface between static and rotating elements and

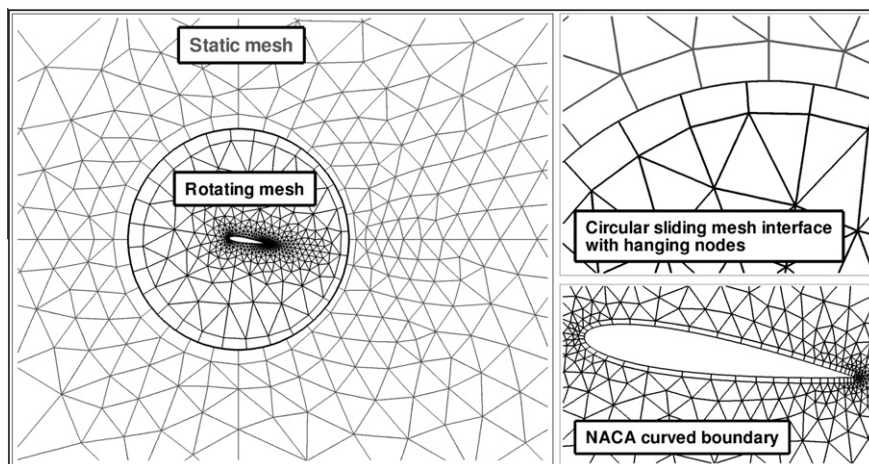


Fig. 1. Mixed triangular–quadrilateral mesh for a symmetric airfoil with curved boundaries and a circular sliding mesh. Static and rotating subdomains are distinguished.

necessitates the use of elements with internal curved edges. In addition, to achieve high order solutions near external boundaries (i.e. walls) curved edged elements are essential. To clarify these concepts before continuing, we depict in Fig. 1 an example of a 2D mesh, where the static and rotating subdomains, the external curved boundary for a symmetric NACA airfoil and the curved sliding mesh interface with the associated hanging nodes have been highlighted.

The present work shows that in the DG context, the *geometric* incompatibility arising from the hanging nodes due to mesh rotation does not cause loss of exponential convergence properties. Further, we show that the *functional* incompatibility originating from the use of triangular and quadrilateral element types (these elements have different numbers of local degrees of freedom) in combination with orthogonal modal basis functions, does not present a problem. Finally, the sliding mesh implementation shows high accuracy when solving the incompressible NS equations in two and three dimensions.

To account for the relative mesh movement of the inner mesh with respect to the static outer mesh, it is advantageous to write the equations for fluid motion in their Arbitrary Lagrangian–Eulerian (ALE) form [11]. The ALE description was first introduced for finite difference methods and unstructured meshes for fluid simulation [12] and subsequently extended to finite elements [13]. A wide explanation of this method can be found in [11].

The ALE approach is generally used for dynamically deforming mesh elements (i.e. arbitrary node movement) and has been widely explored in combination with DG methods to solve hyperbolic type equations including the compressible NS equations (see for example [14–17]). As for the incompressible ALE form of the NS equations with arbitrary mesh movement, work was limited for a long time to conformal h/p spectral discretisations [18]. However, very recent work combines this technique with a DG approach [9].

Deforming element techniques (e.g. ALE for deforming elements) require either generally small body motions or remeshing for large motions that would have otherwise lead to unacceptably element distortions. To avoid these limitations, an appealing approach is provided by combining the ALE approach with the sliding mesh technique. This method is particularly suitable to problems where the mesh movement is known a priori; e.g. rigid body rotation without mesh deformation. We chose to follow this approach, the ALE formulation with sliding meshes for non deforming elements, and summarise some of its advantages below:

- Inertially fixed and rotating objects can be present in the same simulation (see examples in Sections 3.3 and 4).
- No remeshing is required (with its associated computational cost) as mesh elements do not distort, enabling unlimited rotation as opposed to non-sliding or deforming ALE methods in which large distortions and subsequent remeshing need to be handled.
- Subsequently, no projection of the solution into a new mesh is required to advance the solution in time, which is generally a non-conservative process.
- In our approach, no interpolation through the sliding interface is required (as in low order methods) which would introduce large numerical errors, destroying the high order properties of the method.

The ALE approach with sliding mesh interfaces, has been previously studied in the context of h/p conformal spectral methods that require *mortar* techniques for subdomain linking [19]. Recently, a sliding mesh capability has been described for the incompressible NS equations [20], where the authors used *isogeometric analysis* and a low order conformal discretisation (using NURBS) coupled with a special treatment for the sliding curved interface exploiting ideas from DG methods. Purely DG discretisations in conjunction with the sliding mesh approach, have been used for the solution of electromagnetic problems (i.e. Maxwell equations) [21,22]. In [22], it was shown that to solve the Maxwell eigensystem with sliding mesh interfaces, *mortar* techniques are beneficial.

To the authors' knowledge, the work presented herein details the first high order ($order \geq 3$) DG solver with sliding meshes for the solution of the incompressible NS equations. We present a novel approach where non-conformal DG is used in all elements and the curved sliding mesh interface is approximated through an analytical mapping for the description of the circular interface. We show that our approach and implementation does not require of *mortar* type techniques to accurately solve the incompressible NS equations. In addition, to account for curved external boundary surfaces (i.e. walls) representing NACA 4-digit airfoils, we introduce an analytical mapping as is required to obtain smooth solutions for high order techniques. An efficient three dimensional extension of the 2D DG solver is included, which enables 3D flow solutions on rotating geometries that present a geometrical homogeneity in the spanwise (Fourier) direction.

The organisation of the paper is as follows. We firstly introduce, in Section 2, the methodology used to discretise the equations in time and space, with particular emphasis on the required mappings to account for the curved edges (Section 2.2.4), the discretisation and implementation of surface integrals on the curved sliding mesh (Section 2.2.5) and the 3D extension using Fourier series (Section 2.3). Test cases to assess the accuracy, convergence properties and usability of the solver can be found in Sections 3 and 4. Namely, two dimensional flows are verified through the Taylor vortex problem, a static and rotating NACA0015 airfoil and a cross-flow turbine in Section 3. Section 4 concludes with two applications for three dimensional flows: a three bladed cross-flow turbine and a circular cylinder shadowed by a dynamically pitching NACA0012 airfoil.

2. Methodology

We are interested in solving the incompressible NS equations in primitive variable formulation in a unique domain composed of non overlapping static and moving subdomains: $\Omega(t) = \Omega_{sta} \cup \Omega_{rot}(t)$ where $\Omega_{sta} \cap \Omega_{rot}(t) = \Gamma_{sta-rot}$ and $\Gamma_{sta-rot}$

defines the circular sliding mesh interface. The non-dimensional incompressible NS equations in ALE form [11] can be written (in the absence of body forces) as:

$$\frac{\partial \mathbf{u}}{\partial t} + ((\mathbf{u} - \mathbf{w}) \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{\text{Re}} \nabla^2 \mathbf{u}; \quad \nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega(t), \quad (1)$$

where $\mathbf{u} = (u, v)^T$ and $\mathbf{w}$ are non-dimensionalised using the characteristic free stream flow velocity magnitude U and represent the flow and mesh velocities, respectively. Re is the Reynolds number (i.e. $\text{Re} = UL/\nu$ where L is a characteristic length scale and ν is the kinematic viscosity), t represents the dimensionless time and p is the non-dimensional pressure (normalised using upstream dynamic pressure $q = \rho U^2$, where ρ is the fluid density). We distinguish between the static and rotating subdomain meshes (see Fig. 1) by setting:

$$\mathbf{w} = \begin{cases} \mathbf{0} & \forall el \in \Omega_{sta}, \\ \boldsymbol{\varpi} \times \mathbf{x} & \forall el \in \Omega_{rot}(t), \end{cases} \quad (2)$$

where el are the mesh elements in $\Omega(t)$, $\boldsymbol{\varpi} = (0, 0, L\omega/U)$ is the non-dimensional mesh rotational velocity and ω represents the dimensional mesh rotational speed.

To solve the described system, we select to discretise temporally using finite differences (here a dual splitting method), and a high order DG-Fourier method for spatial discretisation. The following Sections 2.1, 2.2 and 2.3 detail these steps.

2.1. Arbitrary Lagrangian–Eulerian temporal discretisation

We perform the temporal discretisation using a second order splitting scheme developed in [23] and studied previously by the authors in conjunction with a DG method for the incompressible NS equations in Eulerian form [10]. The ALE version of this algorithm has been previously used in the context of h/p spectral discretisations [18,19]. Within this scheme the non-linear terms are treated explicitly whilst viscous and pressure terms are handled implicitly. The resulting temporally discretised momentum equation is given by:

$$\frac{\gamma_0 \mathbf{u}^{n+1} - \alpha_0 \mathbf{u}^n - \alpha_1 \mathbf{u}^{n-1}}{\Delta t} = -\nabla p^{n+1} + \frac{1}{\text{Re}} \nabla^2 \mathbf{u}^{n+1} - \beta_0 [((\mathbf{u}^n - \mathbf{w}^n) \cdot \nabla) \mathbf{u}^n] - \beta_1 [((\mathbf{u}^{n-1} - \mathbf{w}^{n-1}) \cdot \nabla) \mathbf{u}^{n-1}]. \quad (3)$$

Using second order Adams–Bashforth values for the coefficients, we set: $\gamma_0 = 3/2$, $\alpha_0 = 2$, $\alpha_1 = -1/2$, $\beta_0 = 2$ and $\beta_1 = -1$. This second order method is not self-starting and we select: $\gamma_0 = \alpha_0 = \beta_0 = 1$ and $\alpha_1 = \beta_1 = 0$ which reduces to the first order explicit forward Euler method for the first time step.

For rigid mesh motion, the mesh movement is known a priori and thus the position of the new nodes can be analytically computed as shown by Eq. (5). The algorithm for the incompressible NS equations in ALE form follows through sequential solution of Eqs. (4)–(7), where Eqs. (6) and (7) are computed using a DG-Fourier variational framework (see Sections 2.2 and 2.3):

$$\tilde{\mathbf{u}} = (\alpha_0/\gamma_0) \mathbf{u}^n + (\alpha_1/\gamma_0) \mathbf{u}^{n-1} - (\beta_0 \Delta t/\gamma_0) [((\mathbf{u}^n - \mathbf{w}^n) \cdot \nabla) \mathbf{u}^n] - (\beta_1 \Delta t/\gamma_0) [((\mathbf{u}^{n-1} - \mathbf{w}^{n-1}) \cdot \nabla) \mathbf{u}^{n-1}], \quad (4)$$

$$\mathbf{x}^{n+1} = \mathbf{R}(\Delta\theta)(\mathbf{x}^n - \mathbf{x}_0) + \mathbf{x}_0, \quad (5)$$

$$-\nabla^2 p^{n+1} = -\frac{\gamma_0}{\Delta t} \nabla \cdot \tilde{\mathbf{u}}, \quad (6)$$

$$-\nabla^2 \mathbf{u}^{n+1} + \frac{\text{Re} \gamma_0}{\Delta t} \mathbf{u}^{n+1} = \text{Re} \left(\frac{\gamma_0}{\Delta t} \tilde{\mathbf{u}} - \nabla p^{n+1} \right), \quad (7)$$

where Δt defines the non-dimensional time step that separates the discrete time levels $n-1$, n and $n+1$ and $\Delta\theta = \Delta t L\omega/U$ is the non-dimensional discrete differential rotation angle, corresponding to the difference between angle at time n (i.e. θ^n) and $n+1$ (i.e. θ^{n+1}). The rotation center for the associated circle is represented by $\mathbf{x}_0$. We further define $\mathbf{R}(\Delta\theta)$ as the rotation matrix:

$$\mathbf{R}(\Delta\theta) = \begin{pmatrix} \cos \Delta\theta & -\sin \Delta\theta \\ \sin \Delta\theta & \cos \Delta\theta \end{pmatrix}. \quad (8)$$

Eq. (6) defines a Poisson equation for the pressure which is closed using the following boundary condition at walls and inflow:

$$\frac{\partial p^{n+1}}{\partial \mathbf{n}} = \mathbf{n} \cdot \frac{\partial \mathbf{u}^{n+1}}{\partial \mathbf{t}} - \beta_0 \mathbf{n} \cdot \left(((\mathbf{u}^n - \mathbf{w}^n) \cdot \nabla) \mathbf{u}^n + \frac{1}{\text{Re}} \nabla \times \boldsymbol{\omega}^n \right) - \beta_1 \mathbf{n} \cdot \left(((\mathbf{u}^{n-1} - \mathbf{w}^{n-1}) \cdot \nabla) \mathbf{u}^{n-1} + \frac{1}{\text{Re}} \nabla \times \boldsymbol{\omega}^{n-1} \right), \quad (9)$$

where $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity. This equation may be derived by taking the dot product of $\mathbf{n}$, the outward normal vector at required boundaries with the momentum equation. Dirichlet conditions are set on the pressure at outflow boundaries Eq. (7) defines a Helmholtz equation for the velocity vector, which is closed using Dirichlet boundary conditions at inflow and walls and Neumann at outflow.

We now particularise the no-slip boundary conditions (i.e. walls) when present in the rotating subdomain $\Omega_{rot}(t)$. Let us introduce the velocity relative to the moving zone: $\mathbf{u}_{rel} = \mathbf{u} - \mathbf{w}$ and note that for rotating no-slip boundary conditions the relative velocity vanishes ($\mathbf{u}_{rel} = \mathbf{0}$). The required Dirichlet condition for velocity becomes $\mathbf{u} = \mathbf{w}$.

The Poisson boundary condition defined in Eq. (9) can be improved by expanding the time derivative of the velocity and considering $\mathbf{u}_{rel} = \mathbf{0}$ at no-slip moving boundaries:

$$\frac{\partial \mathbf{u}^{n+1}}{\partial t} = \frac{\partial (\mathbf{u}_{rel} + \mathbf{w})^{n+1}}{\partial t} = \frac{\partial \mathbf{w}^{n+1}}{\partial t} = \frac{\partial [\boldsymbol{\omega} \times \mathbf{x}]^{n+1}}{\partial t} = \boldsymbol{\omega} \times \frac{\partial \mathbf{x}^{n+1}}{\partial t} = \boldsymbol{\omega} \times \mathbf{w} = \boldsymbol{\omega} \times \boldsymbol{\omega} \times \mathbf{x}^{n+1}, \quad (10)$$

where we have used the definitions: $\mathbf{w} = \boldsymbol{\omega} \times \mathbf{x}$, $\frac{\partial \mathbf{x}^{n+1}}{\partial t} = \mathbf{w}^{n+1}$ and considered constant the rotational speed $\boldsymbol{\omega}^{n+1} = \boldsymbol{\omega}^n = \boldsymbol{\omega}$ (i.e. zero angular acceleration $\frac{\partial \boldsymbol{\omega}}{\partial t} = 0$).

2.2. Two dimensional Discontinuous Galerkin spatial discretisation

The previous temporal scheme requires the spatial solution of non-linear terms Eq. (4) and of elliptic equations Eqs. (6) and (7) at each time step. This section summarises the 2D DG spatial discretisation of these terms.

2.2.1. Preliminaries: spaces and traces

Let $\Omega = \Omega_{sta} \cup \Omega_{rot}$ be a domain in $\mathbb{R}^2$ fixed at an instant in time, with boundaries $\partial\Omega$ (straight or curved) of Dirichlet (Γ_D) or Neumann (Γ_N) type, where $\Gamma_D \cap \Gamma_N = \emptyset$. We introduce a triangular–quadrilateral tessellation $\Omega_h = \{el^{tri}, el^{quad}\}$ of Ω with external boundaries $\partial\Omega_h$ and interior boundaries Γ_h (straight or curved). We further define for each $el \in \Omega_h$ with boundaries ∂el , Γ as the positively oriented, i.e. anticlockwise, single edge of the element boundaries ∂el and the normal vector $\mathbf{n}_\Gamma$ as its unit outward pointing normal. In addition, let us define $\mathbf{n}_{\Gamma-local}$ as the pointwise unit outward pointing normal, which is required when considering curved edges.

We consider the discontinuous approximate space $D_k(\Omega_h) = D_k^{tri}(\Omega_h) \cup D_k^{quad}(\Omega_h)$ with:

$$\begin{aligned} D_k^{tri}(\Omega_h) &= \left\{ v_h \in L^2(\Omega) : \forall el^{tri} \in \Omega_h, v_h|_{el^{tri}} \in \mathbb{P}_k(el^{tri}) \right\}, \\ D_k^{quad}(\Omega_h) &= \left\{ v_h \in L^2(\Omega) : \forall el^{quad} \in \Omega_h, v_h|_{el^{quad}} \in \mathbb{Q}_k(el^{quad}) \right\} \end{aligned} \quad (11)$$

and their vector version $[D_k^{tri}(\Omega_h)]^2$, $[D_k^{quad}(\Omega_h)]^2$ and $[D_k(\Omega_h)]^2 = [D_k^{tri}(\Omega_h)]^2 \cup [D_k^{quad}(\Omega_h)]^2$, where $\mathbb{P}_k(el^{tri})$ and $\mathbb{Q}_k(el^{quad})$ denote the space of polynomials of order less than or equal to k for triangular and quadrilateral elements respectively (i.e. we will only consider same polynomial orders for triangles and quadrilaterals) and equal order elements for velocity and pressure.

We also define the jump $[[\bullet]]$ and average $\{\{\bullet\}\}$ across the interface between neighbouring elements el_1 and el_2 , arising from the discontinuous discretisation:

$$[[\bullet]] = (\bullet|_{el_1}) - (\bullet|_{el_2}), \{\{\bullet\}\} = \frac{1}{2}(\bullet|_{el_1}) + \frac{1}{2}(\bullet|_{el_2}), \quad \forall \Gamma = \partial el_1 \cap \partial el_2 \quad (12)$$

and at boundary edges as: $[[\bullet]] = \{\{\bullet\}\} = (\bullet|_{el_1}), \quad \forall \Gamma = \partial el_1 \cap \partial\Omega_h$.

2.2.2. Spatial discretisation of the non-linear terms

The convective form of the non-linear terms in the Navier–Stokes equations Eq. (1) are expressed under discretisation in their skew-symmetric version. The following formulation (under its weak setting) is proposed: find $\mathbf{u}_h \in [D_k(\Omega_h)]^2$ such that:

$$\int_{\Omega_h} (((\mathbf{u}_h - \mathbf{w}) \cdot \nabla) \mathbf{u}_h) \cdot \mathbf{v}_h d\mathbf{x} = \sum_{el \in \Omega_h} \int_{el} (((\mathbf{u}_h - \mathbf{w}) \cdot \nabla) \mathbf{u}_h) \cdot \mathbf{v}_h d\mathbf{x} + 1/2 \sum_{el \in \Omega_h} \int_{el} ((\nabla \cdot (\mathbf{u}_h - \mathbf{w})) \mathbf{u}_h) \cdot \mathbf{v}_h d\mathbf{x} \quad (13)$$

for all $\mathbf{v}_h \in [D_k(\Omega_h)]^2$. Let us note that for rigid body motion $\nabla \cdot \mathbf{w} = 0$. Previous developments on DG methods to solve the incompressible NS equations have required the use of numerical fluxes for the non-linear terms discretisation. Examples can be found in [6,24] where Lax–Friedrich DG variants were used or in [4], where a DG modification of the Lesaint–Raviart approximation was incorporated. We note that the present implementation does not require upwinding or flux enforcement and only volume terms are retained, which reduces the computational cost for the evaluation of the non-linear terms. This discretisation is tightly related to the selected time advancement procedure and has a threefold justification:

1. We are only interested in continuous solutions where viscosity dominates and there is lack of hyperbolicity (e.g. flow discontinuities, shocks).
2. Within the splitting scheme used, the non-linear terms act as forcing terms in the Poisson and Helmholtz equations (Eqs. (6) and (7)). If surface integrals (e.g. upwinding) with jumps and averages are added to the non-linear terms Eq. (13), these will have a small contribution when compared to the volume terms. Furthermore, as the polynomial order increases the numerical solution tends to be continuous (for a viscosity dominant flow) hence reducing the contribution of the jumps (assuming a consistent and conservative formulation for the numerical flux). Overall, the forcing terms will effectively be unchanged and so will the solution.

3. From the stability point of view of the temporal scheme, positivity is not required since the non-linear terms are treated explicitly.

Let us note that the skew-symmetric form has been selected since it minimises aliasing errors [23]. In addition, in our implementation we use over-integration, as detailed in [25] for conformal h/p spectral discretisations, since it has been shown that aliasing errors are further minimised.

The explicit treatment of the non-linear terms introduces a Courant–Friedrichs–Lewy (CFL) type restriction on the time step. The CFL estimate for high order spatial methods leads to $\Delta t < \Delta t_{\text{CFL}} = \mathcal{O}(h/Uk^2)$ as shown in [1] for an advection model problem, where h is the mesh element size, U the maximum velocity and k represents the polynomial order. For the ALE formulation, this estimate becomes: $\Delta t < \Delta t_{\text{CFL}} = \mathcal{O}(h/|\mathbf{u} - \mathbf{w}|k^2)$, where $\mathbf{u}$ and $\mathbf{w}$ are the flow and mesh velocities. If we consider the limiting case $|\mathbf{w}|_{\text{max}} = R\omega$ (i.e. the sliding interface), where R defines the radius of the circle associated to the sliding interface and ω the rotational speed, we can identify the most stringent case which occurs when $\mathbf{w}$ and $\mathbf{u}$ (with magnitude $|\mathbf{u}| = U$) have opposite directions: $\Delta t_{\text{CFL}} = \mathcal{O}(h/(U + R\omega)k^2)$.

2.2.3. The DG-SIPG spatial discretisation for elliptic equations

The described temporal splitting method requires the solution of purely elliptic equations (Poisson equation (6) and Helmholtz equation (7)). Among the various existing DG formulations, we select the Symmetric Interior Penalty Galerkin (SIPG) method to spatially discretise these equations. This formulation has the advantage over other Interior Penalty forms (see [4]) to lead to a symmetric positive definite system allowing the use of fast iterative solvers. The details of the development and derivation are omitted but can be found in [4]. We introduce here the general formulation for the continuous elliptic problem which encompasses the Poisson and Helmholtz equations:

$$\begin{aligned} -\Delta u + \alpha u &= g \quad \text{in } \Omega, \\ u &= L_D \quad \text{on } \Gamma_D, \\ \nabla u \cdot \mathbf{n} &= L_N \quad \text{on } \Gamma_N, \end{aligned} \quad (14)$$

where $u \in H^1(\Omega)$ is a scalar solution (but extension to the vector formulation is direct), $\sqrt{\alpha}$ is real and represents the wave number for the Helmholtz equation, $\mathbf{n}$ is the outward pointing unit normal vector on $\partial\Omega$, $g \in L^2(\Omega)$ is the forcing term, and $L_D \in H^{1/2}(\Gamma_D)$ and $L_N \in L^2(\Gamma_N)$ represent the boundary conditions. To find the weak solution of the continuous problem represented by Eq. (14) using the DG-SIPG method, we introduce the bilinear $a(\bullet, \bullet)$ and linear $\ell(\bullet)$ forms associated with the discrete finite element variational formulation arising from the weak form of the continuous problem defined by Eq. (15). The bilinear form is defined as:

$$\begin{aligned} a(u_h, v_h) &= \sum_{el \in \Omega_h} \int_{el} (\nabla u_h \cdot \nabla v_h) d\mathbf{x} + \sum_{el \in \Omega_h} \int_{el} (\alpha u_h v_h) d\mathbf{x} - \sum_{\Gamma \in \mathcal{F}_h \cup \Gamma_D} \int_{\Gamma} \{\{\nabla u_h\}\} \cdot \mathbf{n}_{\Gamma} [[v_h]] ds \\ &\quad - \sum_{\Gamma \in \mathcal{F}_h \cup \Gamma_D} \int_{\Gamma} \{\{\nabla v_h\}\} \cdot \mathbf{n}_{\Gamma} [[u_h]] ds + \sum_{\Gamma \in \mathcal{F}_h \cup \Gamma_D} \frac{\sigma}{|trace|^\beta} \int_{\Gamma} [[u_h]] [[v_h]] ds \end{aligned} \quad (15)$$

and the following linear form:

$$\ell(v_h) = \sum_{el \in \Omega_h} \int_{el} g v_h d\mathbf{x} - \sum_{\Gamma \in \mathcal{F}_h} \int_{\Gamma} \left(\nabla v_h \cdot \mathbf{n}_{\Gamma} - \frac{\sigma}{|trace|^\beta} v_h \right) L_D ds + \sum_{\Gamma \in \Gamma_N} \int_{\Gamma} v_h L_N ds \quad (16)$$

for all $v_h \in D_k(\Omega_h)$, where σ and β are positive real parameters and $|trace|$ is defined as the length of the element edge in 2D. Then, the DG-SIPG approximate solution reduces to finding $u_h \in D_k(\Omega_h)$ provided by:

$$a(u_h, v_h) = \ell(v_h), \quad \forall v_h \in D_k(\Omega_h). \quad (17)$$

The combination of σ and β in Eqs. (15) and (16), defines the penalty parameter and has to be chosen to be large enough to enforce coercivity of the bilinear form [4]. We use $\sigma = 3k(k+1)$ and $\beta = 1$ for triangles (see [10] for details on the derivation). For quadrilateral elements, we also use $\sigma = 3k(k+1)$ as a conservative estimate (generally $\sigma \approx k^2$) of the penalty parameter.

2.2.4. Mapping from computational to physical space

In this section we introduce the necessary mappings to relate computational to physical space for the three types of elements considered: straight sided triangular elements, straight sided quadrilateral elements and quadrilateral elements with one curved edge.

The non-linear terms, Eq. (13), and the SIPG weak form, Eqs. (15) and (16), require computation of volume and surface integrals. To perform these integrations, we consider mappings Ψ from the reference element I_{el} in computational space $\xi = (\xi_1, \xi_2)$ to the element el in physical space $\mathbf{x} = (x, y)$ as $I_{el} := \Psi^{-1}(el)$ and define the determinant of the Jacobian $|\mathbf{J}_\Psi|$ for the transformation. We also account for the transformation of the edges Γ of element el in physical space to the mapped edge in the computational space: $\Gamma = \mathcal{C}[-1, 1])$ and define the norm of the differential parametrisation (i.e. differential length) of $\mathcal{C}(t)$ as $|\mathcal{C}'(t)|$. Using these definitions, we can equate the volume and surface integrals for a general polynomial function $f(\mathbf{x})$ defined in physical space to their equivalent form in computational space:

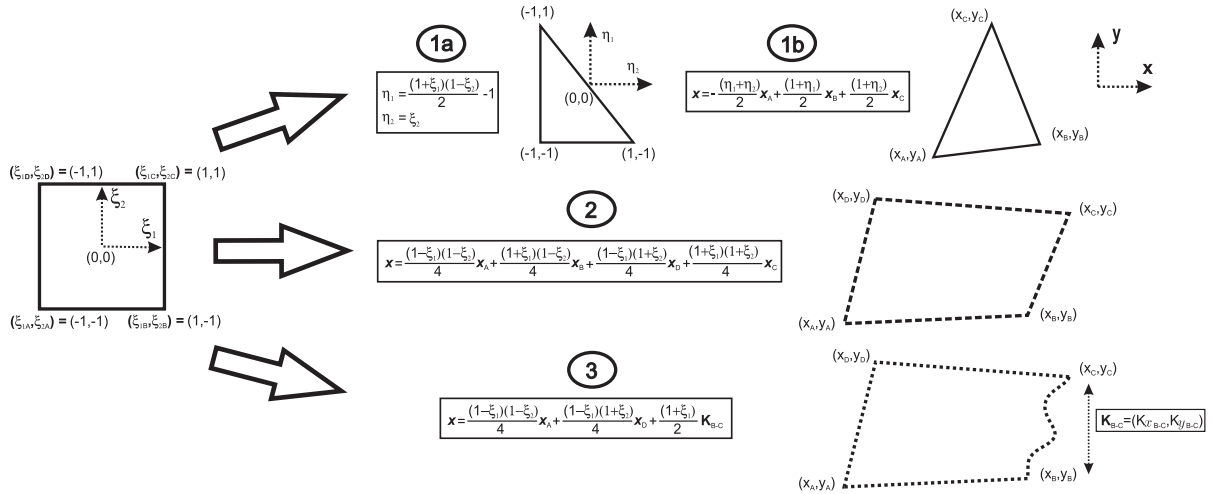


Fig. 2. Mapping from computational space to physical space for straight sided triangular (map 1a and 1b), straight sided quadrilateral (map 2) and quadrilateral elements with one curved edge (map 3).

$$\int_{el} f(\mathbf{x}) d\mathbf{x} = \int_{el} f(\xi) |\mathbf{J}_{\Psi}| d\xi, \quad (18)$$

$$\int_{\Gamma} f(\mathbf{x}) ds = \int_{-1}^1 f(\mathbf{C}(t)) |\mathbf{J}_{\mathbf{C}}(t)| dt. \quad (19)$$

Three different mappings Ψ are considered and depicted in Fig. 2:

- **1a and 1b – Triangular map:** (1a) standard square (ξ_1, ξ_2) to standard triangle (η_1, η_2) and (1b) standard triangle to arbitrary triangle (x, y) (see [1] for details).
- **2 – Quadrilateral map:** standard square (ξ_1, ξ_2) to arbitrary quadrilateral (x, y) (see [26] for details).
- **3 – Curved-quadrilateral map:** standard square (ξ_1, ξ_2) to arbitrary quadrilateral with one curved edge (x, y) (see [26] for details).

The mappings defined in Fig. 2 lead to a constant determinant of the Jacobian $|\mathbf{J}_{\Psi}|$ for straight sided triangular elements and quadrilateral elements with straight parallel edges. However, this determinant is not constant for general quadrilaterals elements. The straight sided triangular elements map is composed by two transformations (see [1] for details) and requires the use of the chain rule for differentiation.

In addition to the determinant of the Jacobian $|\mathbf{J}_{\Psi}|$ and the norm of the differential parametrisation $|\mathbf{J}_{\mathbf{C}}(t)|$ required to evaluate volume and surface integrals (Eqs. (18) and (19)), partial derivatives from physical to computational space (i.e. gradients) and edge normals are also required to compute the non-linear terms Eq. (13) and the SIPG weak form Eqs. (15) and (16). The mappings defined in Fig. 2 allow for the computation of these metrics.

The *Curved-quadrilateral map-3* is necessary to account for external curved boundaries and the interior curved sliding interface. The general approach used in high order methods, to account for curved edged elements, is to use *iso-parametric* mapping. This technique approximates the curved edge using polynomial functions (or splines, NURBS) which have the same degree that the polynomials used within the element to calculate the flow solution [1]. Then the approximated curve is used to compute the required metrics (i.e. $|\mathbf{J}_{\Psi}|$, $|\mathbf{J}_{\mathbf{C}}(t)|$, gradients and normals). Our approach is somewhat different in the sense that it does not require to fit a polynomial to the non-polynomial curve, defining the curved edge, to compute the required metrics from the fitted polynomial. Instead, the analytical expression that defines the curved edges is used to compute the required metrics (at quadrature points). Conceptually, this technique is similar to the NURBS-Enhanced FEM [27] approach since both allow for decoupling of the geometry (and associated metrics) with respect to the polynomial space required to approximate the flow solution. Our approach can be seen as less general, but it can be argued that an analytical expression of the geometry can always be obtained (e.g. by fitting a polynomial of an arbitrary order).

For the curved mapping with one curved edge (*Curved-quadrilateral map-3*), we only consider the edge B–C being curved which corresponds to the edge $\xi = (\xi_1 = 1, \xi_2)$ in the computational space as shown in Fig. 2. This can be ensured by the appropriate node numbering when generating the mesh (but generalisation to other edges is direct). The function $\mathbf{K}_{B-C}(\xi_2) = (K_{x_{B-C}}(\xi_2), K_{y_{B-C}}(\xi_2))$ in Fig. 2 defines the shape of the curved edge. Namely, in this work we have considered two types of curved edges: circular arcs and NACA 4-digit edges.

The circular parametrisation which was previously introduced in [1], is defined as:

$$\begin{aligned} K_{x_{B-C}}^{CIRC}(\xi_2) &= R \cos \left(\frac{1-\xi_2}{2} \theta_B + \frac{1+\xi_2}{2} \theta_C \right), \\ K_{y_{B-C}}^{CIRC}(\xi_2) &= R \sin \left(\frac{1-\xi_2}{2} \theta_B + \frac{1+\xi_2}{2} \theta_C \right), \end{aligned} \quad (20)$$

where $R = R_B = R_C$ and θ_B, θ_C are the polar-coordinates of nodes B and C with respect to an origin at the centre of curvature. This parametrisation is used hereafter to account for the circular sliding interface and can be used to define circular external boundaries. It can be shown that the circular parametrisation $K_{B-C}^{CIRC}(\xi_2)$ provides a constant determinant of the Jacobian $|\mathbf{J}_\Psi|$ provided $\theta_A = \theta_B, \theta_C = \theta_D$ and $R_B = R_C, R_A = R_D$. In our implementation however, we retain the general case where the determinant of the Jacobian is not constant.

For the NACA 4-digit airfoils [28], the analytical expression for the airfoil can be used to define the mapping required to approximate curved external boundaries:

$$\begin{aligned} K_{x_{B-C}}^{NACA}(\xi_2) &= \frac{1-\xi_2}{2} x_B + \frac{1+\xi_2}{2} x_C, \\ K_{y_{B-C}}^{NACA}(\xi_2) &= \pm \frac{\zeta}{0.2} \left(\tau_0 \sqrt{K_{x_{B-C}}^{NACA}(\xi_2)} - \tau_1 [K_{x_{B-C}}^{NACA}(\xi_2)] - \tau_2 [K_{x_{B-C}}^{NACA}(\xi_2)]^2 - \tau_3 [K_{x_{B-C}}^{NACA}(\xi_2)]^3 - \tau_4 [K_{x_{B-C}}^{NACA}(\xi_2)]^4 \right), \end{aligned} \quad (21)$$

where ζ is the airfoil maximum thickness (with respect to unit chord $c = 1$) and $\tau_0 = 0.29690$, $\tau_1 = 0.12600$, $\tau_2 = 0.35160$, $\tau_3 = 0.28430$ and $\tau_4 = 0.20250$ are the NACA constants required to define the profile geometry for symmetric NACA airfoils. The airfoil nose definition requires a circular mapping with radius $r_{nose} = 1.1019\zeta^2$ and we use Eq. (20). The transition from the analytic nose and airfoil body formulations is not particularly smooth, and we select this transition to occur at $x = 0.002$ (instead of the value $x = 0.005$ suggested in [28]) which provides a smooth enough geometric transition for the tested airfoil ($\zeta = 12\%$ and 15%). In all cases, we extend the NACA blunt trailing edge with straight edges to obtain a sharp trailing edge, which extends the actual chord to $c \approx 1.01$. This mapping is defined for $0 < x < 1$ but can be generalised to arbitrary values when combined with translation and rotation.

The required metrics (i.e. $|\mathbf{J}_\Psi|, |\mathbf{J}_\mathbf{C}(t)|$, gradients and normals) can be analytically obtained through the defined mappings, using the chain rule when necessary, to differentiate physical to computational coordinates.

In all cases we use Gauss–Legendre quadrature as it has been shown in [29] that this is more efficient than other quadrature rules (e.g. trapezoidal, Romberg's rule) when dealing with non-polynomial functions. We also ensure that enough quadrature points are used for the correct evaluation of volume and surface integrals.

We finally note that to calculate surface integrals on external curved edges using the defined mappings it suffices to replace $\mathbf{C}(t = \xi_2) = \mathbf{K}_{B-C}(\xi_2)$ in Eq. (19). The flux treatment for curved edges lying on the sliding mesh interface is presented in the next section.

2.2.5. Sliding mesh implementation

Due to the inner domain rotation relative to the outer static mesh, there is, for arbitrary rotation, the creation of hanging nodes and an associated geometric incompatibility. We will see in what follows that this incompatibility does not represent a problem in the DG context.

As detailed in Section 2.1, we require the solution of elliptic problems and have selected the DG-SIPG method for their spatial discretisation. When implementing the sliding mesh technique, it becomes necessary to revisit the bilinear form defined in Section 2.2.3, and in particular the last three terms in Eq. (15), involving surface integrals. Expanding the average and jump terms (i.e. $\{\{\bullet\}\}, [\![\bullet]\!]$) in the surface integrals and replacing the approximate solution by its expansion (see following Section 2.2.6, Eq. (28) for details), we obtain: $\int_\Gamma t_{ij} ds = \int_\Gamma t_{ij}^{el_\ell} ds$, where $i = 1$ to $LDOF$ is the test function index of v_h , $j = 1$ to $LDOF$ is the basis function index of u_h and ℓ, χ denote the elements sharing the edge Γ . We recast these contributions as:

$$\int_\Gamma t_{ij}^{el_\ell} ds = \gamma \frac{1}{2} \int_\Gamma \nabla \phi_j^{el_\ell} \cdot \mathbf{n}_{\Gamma-local} \phi_i^{el_\ell} ds + \delta \frac{1}{2} \int_\Gamma \phi_j^{el_\ell} \nabla \phi_i^{el_\ell} \cdot \mathbf{n}_{\Gamma-local} ds + \rho \frac{\sigma}{|trace|} \int_\Gamma \phi_j^{el_\ell} \phi_i^{el_\ell} ds, \quad (22)$$

where $\mathbf{n}_{\Gamma-local}$ defines the local normal vector (i.e. $\mathbf{n}_{\Gamma-local} = \mathbf{n}_\Gamma$ for straight edges) and γ, δ and ρ are defined as follows:

$$\gamma = \begin{cases} 1; & \ell \neq 1, \\ -1; & \ell = 1, \end{cases} \quad \delta = \begin{cases} 1; & \chi \neq 1, \\ -1; & \chi = 1, \end{cases} \quad \rho = \begin{cases} 1; & \ell = \chi, \\ -1; & \ell \neq \chi. \end{cases} \quad (23)$$

We note that by performing summation over $\ell, \chi = 1, 2$, we retrieve $\int_\Gamma t_{ij} ds = \int_\Gamma (t_{ij}^{11} + t_{ij}^{22} + t_{ij}^{12} + t_{ij}^{21}) ds$, the four flux contributions representative of two unique elements el_1 and el_2 .

In what follows, we consider that the elements sharing the sliding mesh interface have one curved edge and a unique hanging node per element interface. Furthermore, we consider, without loss of generality, that the hanging node lies on the edge $\xi = (\xi_1 = 1, \xi_2)$ in the computational space (i.e. edge B–C in physical space). A schematic of three elements with curved edges sharing the sliding mesh interface and the resulting hanging node is depicted in Fig. 3, where the connectivity between elements in both physical and computational space is detailed. The circular edge mapping previously defined in Eq. (20) is used to describe the curved elements sharing the circular sliding mesh interface. The flux contributions defined in Eq. (22) over the curved interface (i.e. $\Gamma \in \Gamma_{sta-rot}$) have to take into account the hanging node and that the edge is split in two segments: Γ_{12} that connects element el_1 and el_2 and Γ_{13} connecting el_1 to el_3 . We need thus to extend these surface integrals

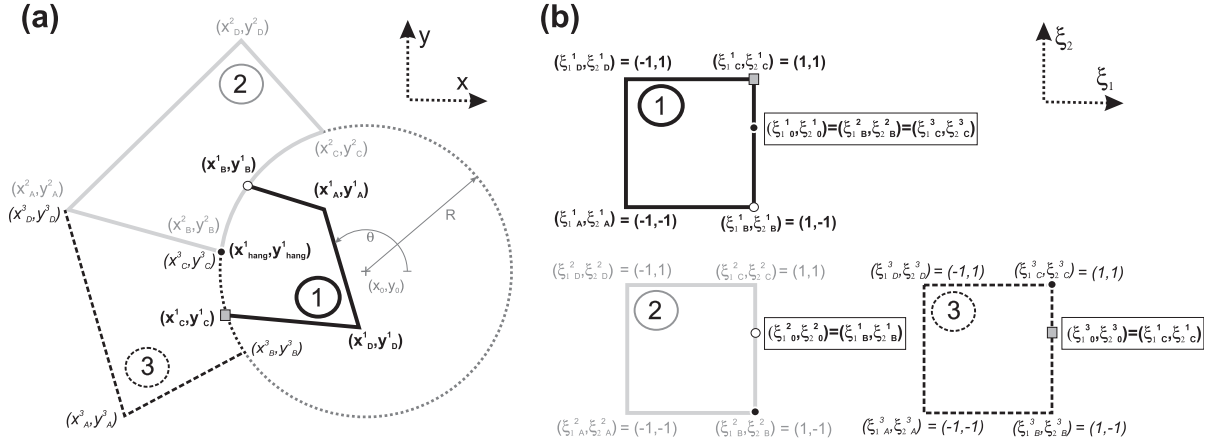


Fig. 3. Sliding mesh connectivity scheme showing three elements and the shared interfaces with a hanging node in (a) the physical space and (b) the computational space.

defined for a single edge (i.e. only shared by two elements) to account for the interface including a hanging node $\Gamma = \Gamma_{12} \cup \Gamma_{13}$. Before proceeding, we define the norm of the differential parametrisation (or differential length) $\|\mathbf{J}_C(t)\|$ introduced in Eq. (19) that defines the surface integral in computational space. Recalling the definition of the circular mapping introduced in Eq. (20) and setting $\mathbf{C}(t) = \mathbf{K}_{B-C}^{CIRC}(\xi_2 = h(t))$, where $h(t)$ represents an linear function to be subsequently defined; we introduce:

$$\|\mathbf{J}_{K_{B-C}^{CIRC}}(t)\| = \sqrt{\left(\frac{\partial \mathbf{K}_{B-C}^{CIRC}(\xi_2)}{\partial \xi_2} \frac{\partial \xi_2}{\partial t}\right)^2 + \left(\frac{\partial \mathbf{K}_{B-C}^{CIRC}(\xi_2)}{\partial \xi_2} \frac{\partial \xi_2}{\partial t}\right)^2} dt. \quad (24)$$

For edges where no hanging nodes are present, we retrieve $h(t) = t$ (i.e. $\xi_2 = t$). We consider the internal element el_1 as the reference element and follow the notation introduced in Fig. 3. Combining Eqs. (19), (22) and (24) we obtain:

$$\begin{aligned} \int_{\Gamma \in \Gamma_{sta-rot}} t_{ij}^{el_i el_\chi}(\mathbf{x}) ds &= \int_{\Gamma_{12}} t_{ij}^{el_i el_\chi}(\mathbf{x}) ds_{12} + \int_{\Gamma_{13}} t_{ij}^{el_i el_\chi}(\mathbf{x}) ds_{13} = \int_{x_B^1 y_B^1}^{x_{hang}^1 y_{hang}^1} t_{ij}^{el_i el_\chi}(\mathbf{x}) ds_{12} + \int_{x_{hang}^1 y_{hang}^1}^{x_C^1 y_C^1} t_{ij}^{el_i el_\chi}(\mathbf{x}) ds_{13} \\ &= \int_{-1}^1 t_{ij}^{el_i el_\chi}(\mathbf{K}_{B-C}^{CIRC}(t_1)) \|\mathbf{J}_{K_{B-C}^{CIRC}}(t_1)\| dt_1 + \int_{-1}^1 t_{ij}^{el_i el_\chi}(\mathbf{K}_{B-C}^{CIRC}(t_2)) \|\mathbf{J}_{K_{B-C}^{CIRC}}(t_2)\| dt_2, \end{aligned} \quad (25)$$

where $-1 \leq t_1, t_2 \leq 1$ are the parametrisations for Γ_{12} and Γ_{13} respectively and $\ell, \chi \neq 3$ in Γ_{12} and $\ell, \chi \neq 2$ in Γ_{13} . We can relate this parametrisation to the computational space by introducing the following intermediate mappings for $h(t)$:

$$\begin{aligned} \xi_2 &= h_{12}(t_1) = \frac{1-t_1}{2}(-1) + \frac{1+t_1}{2}\xi_{20}^1; \quad \text{on } \Gamma_{12}, \\ \xi_2 &= h_{13}(t_2) = \frac{1-t_2}{2}\xi_{20}^1 + \frac{1+t_2}{2}(1); \quad \text{on } \Gamma_{13}, \end{aligned} \quad (26)$$

where we have defined the hanging node on the computational space that maps to the hanging node in physical space (see Fig. 3) as: $(\xi_{10}^1, \xi_{20}^1) \leftrightarrow (x_{hang}^1, y_{hang}^1)$ and note that $\xi_{10}^1 = 1$ (since $\xi_1 = 1$) and $\xi_{20}^1 = 2 \frac{(\theta_{hang}^1 - \theta_B^1)}{(\theta_C^1 - \theta_B^1)} - 1$, with $\theta_{hang}^1, \theta_B^1, \theta_C^1$ the corresponding angles for the reference element in physical space. We can differentiate the map for the circular edge Eq. (20) and the intermediate mapping Eq. (26) to obtain the required definition of the norm of the differential parametrisation $\|\mathbf{J}_{K_{B-C}^{CIRC}}(t)\|$, Eq. (24). Substituting this expression into Eq. (25), we derive the necessary definition for the fluxes across the curved interfaces with a hanging node:

$$\int_{\Gamma \in \Gamma_{sta-rot}} t_{ij}^{el_i el_\chi}(\mathbf{x}) ds = \frac{\theta_C - \theta_B}{4} R \left[(\xi_{20}^1 + 1) \int_{-1}^1 t_{ij}^{el_i el_\chi}(\mathbf{K}_{B-C}^{CIRC}(t_1)) dt_1 + (1 - \xi_{20}^1) \int_{-1}^1 t_{ij}^{el_i el_\chi}(\mathbf{K}_{B-C}^{CIRC}(t_2)) dt_2 \right], \quad (27)$$

with $i, j = 1: LDOF, \ell, \chi = 1, 2, 3$, but $\ell, \chi \neq 3$ in the first integral and $\ell, \chi \neq 2$ in the second. These integrals can now be evaluated using Gaussian quadrature.

The penalty term in Eq. (22) requires a definition for the $|trace|$ and we select: $|trace|_{12} = |trace|_{13} = |\theta_C - \theta_B|R$.

2.2.6. 2D modal basis functions

Within each element el , the approximate solution, u_h is expanded using modal basis functions $\phi_i(x, y)$ as:

$$u_h(x, y) = \sum_{i=1}^{LDOF} u_i \phi_i(x, y), \quad (28)$$

where $x, y \in el$ and $LDOF$ is the local number of degrees of freedom. We select a hierarchical tensorial and orthonormal (with respect to the L^2 -inner product) expansion bases (see [1] for details) defined in both physical and computational space (see nomenclature in Fig. 2) as:

$$\begin{aligned}\phi_i^{quad}(x, y) &= \phi_{pq}^{quad}(x, y) = \phi_p^{quad^A}(\xi_1) \phi_q^{quad^B}(\xi_2), \\ \phi_i^{tri}(x, y) &= \phi_{pq}^{tri}(x, y) = \phi_p^{tri^A}(\eta_1, \eta_2) = \phi_p^{tri^A}(\xi_1) \phi_{pq}^{tri^B}(\xi_2),\end{aligned}\quad (29)$$

where p, q denote the different components of the tensorial expansion. We express the basis functions for quadrilateral and triangular elements in the computational domain (ξ_1, ξ_2) in terms of Jacobi polynomials:

$$\begin{aligned}\phi_p^{quad^A}(\xi_1) &= \sqrt{\frac{2p+1}{2}} P_p^{(0,0)}(\xi_1), \quad \phi_q^{quad^B}(\xi_2) = \sqrt{\frac{2q+1}{2}} P_q^{(0,0)}(\xi_2); \\ \phi_p^{tri^A}(\xi_1) &= \sqrt{\frac{2p+1}{2}} P_p^{(0,0)}(\xi_1), \quad \phi_{pq}^{tri^B}(\xi_2) = \sqrt{p+q+1} \left(\frac{1-\xi_2}{2}\right)^p P_q^{(2p+1,0)}(\xi_2),\end{aligned}\quad (30)$$

where $P_p^{(\alpha,\beta)}(z)$ is the p th order Jacobi polynomial. For quadrilateral elements, $0 \leq p, q \leq k$ and $LDOF_{quad} = (k+1)^2$, given a polynomial order k . For triangular elements $0 \leq p, q$ and $p+q \leq k$ and $LDOF_{tri} = (k+1)(k+2)/2$. It can be seen that since $LDOF_{quad} \neq LDOF_{tri}$ and no special treatment is performed at element edges, functional incompatibilities may arise for adjacent triangular and quadrilateral elements, however, the result sections show that these incompatibilities do not present a problem in the DG context.

For both element types, the bases have been normalised to provide mass matrices that are the identity matrix for triangles and quadrilaterals with parallel edges (i.e. constant determinant of the Jacobians): $\mathbf{M} = \mathbf{J}_\psi \mathbf{Id}$. When considering a non-constant determinant of the Jacobian (e.g. quadrilateral elements with a curved edge) in conjunction with the defined basis functions, a symmetric mass matrix (i.e. $\mathbf{M} = \mathbf{M}^T$) is obtained, which substantially reduces the costs of computing the matrix and subsequent operations involving the mass matrix, relative to costs for a full mass matrix.

2.3. Three dimensional Fourier extension

The selected tensorial representation of the mappings from physical to computational space (Section 2.2.4) in combination with tensorial orthogonal basis functions (Section 2.2.6) allow for 3D extensions of the method without further modifications. In this section, we describe a 3D extension where the two dimensional Discontinuous Galerkin capabilities presented in the previous sections are extended to account for three dimensional flow features using Fourier series. The selected Fourier extension enables the use of the Fast Fourier Transform (FFT) algorithm and provides an efficient 3D solver for problems involving spanwise geometric homogeneity.

The use of Fourier series, to efficiently extend two dimensional codes and account for three dimensional flows, has been previously reported for h/p conformal spectral methods (e.g. [1,30,31]). To the authors' knowledge, this type of extension has not been used together with DG discretisations. However, this technique leads to a series of 2D DG linear solves that are coupled through the non-linear terms (see details below), in a similar manner to conformal implementations. This feature renders these Fourier extensions similar.

Let us introduce the three dimensional DG-Fourier velocities and pressure as:

$$\mathbf{u}(x, y, z) = \sum_{m=-M/2}^{M/2-1} \mathbf{u}_m(x, y) e^{i\beta m z}; \quad p(x, y, z) = \sum_{m=-M/2}^{M/2-1} p_m(x, y) e^{i\beta m z}, \quad (31)$$

where $i = \sqrt{-1}$, $\beta = 2\pi/L_z$ represents the wavenumber in the z -direction and L_z defines the length of the domain in the periodic z -direction. In addition, M defines the number of planes in physical space (and the number of modes in the Fourier space) and m denotes the m th mode of the Fourier transformation. Let us note that $\mathbf{u}_m(x, y)$ and $p_m(x, y)$ have been previously defined through Eq. (28) and represent the x - y plane DG solution for the m th Fourier mode.

Applying the Fourier transform to the momentum and continuity equations in ALE form Eq. (1) and using its linearity property, one obtains:

$$\sum_{m=-M/2}^{M/2-1} \left\{ \frac{\partial \mathbf{u}_m}{\partial t} + \hat{\nabla}_m p_m - \frac{1}{\text{Re}} \hat{\nabla}_m^2 \mathbf{u}_m + \text{FFT}_m \{ ((\mathbf{u} - \mathbf{w}) \cdot \nabla) \mathbf{u} \} \right\} e^{i\beta m z} = 0; \quad \sum_{m=-M/2}^{M/2-1} \left\{ \hat{\nabla}_m \cdot \mathbf{u}_m \right\} e^{i\beta m z} = 0 \quad \text{in } \Omega(t). \quad (32)$$

where $\text{FFT}_m\{\bullet\}$ represent the m th mode of the Fourier transform of the argument and the Fourier operators $\hat{\bullet}$ are defined as:

$$\hat{\nabla}_m = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, i\beta m \right)^T = (\nabla_{2D}, i\beta m)^T, \quad (33)$$

$$\hat{\nabla}_m^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} - \beta^2 m^2 \right) = \nabla_{2D}^2 - \beta^2 m^2, \quad (34)$$

where ∇_{2D} and ∇_{2D}^2 are the two dimensional gradient (nabla) and Laplacian operators, respectively.

The system described by Eq. (32) is equivalent to solving an independent equation for each mode m :

$$\frac{\partial \mathbf{u}_m}{\partial t} + FFT_m\{((\mathbf{u} - \mathbf{w}) \cdot \nabla) \mathbf{u}\} = -\hat{\nabla}_m p_m + \frac{1}{\text{Re}} \hat{\nabla}_m^2 \mathbf{u}_m; \hat{\nabla}_m \cdot \mathbf{u}_m = 0 \quad \text{in } \Omega_m(t). \quad (35)$$

The coupling between modes is ensured by the non-linear terms which are computed in physical space. It should be noted that this Fourier extension only requires real to complex transforms and hence only $M/2 + 1$ modes need to be computed at each time step.

Finally, the temporal discretisation algorithm for the 3D incompressible NS equations in ALE form described in Section 2.1 is extended to:

$$\tilde{\mathbf{u}}_m = (\alpha_0/\gamma_0) \mathbf{u}_m^n + (\alpha_1/\gamma_0) \mathbf{u}_m^{n-1} - (\beta_0 \Delta t/\gamma_0) FFT_m\{((\mathbf{u}^n - \mathbf{w}^n) \cdot \nabla) \mathbf{u}^n\} - (\beta_1 \Delta t/\gamma_0) FFT_m\{((\mathbf{u}^{n-1} - \mathbf{w}^{n-1}) \cdot \nabla) \mathbf{u}^{n-1}\}, \quad (36)$$

$$\mathbf{x}^{n+1} = \mathbf{R}(\Delta\theta)(\mathbf{x}^n - \mathbf{x}_0) + \mathbf{x}_0, \quad (37)$$

$$(-\nabla_{2D}^2 + \beta^2 m^2) p_m^{n+1} = -\frac{\gamma_0}{\Delta t} \hat{\nabla}_m \cdot \tilde{\mathbf{u}}_m, \quad (38)$$

$$\left(-\nabla_{2D}^2 + \beta^2 m^2 + \frac{\text{Re}\gamma_0}{\Delta t}\right) \mathbf{u}_m^{n+1} = \text{Re}\left(\frac{\gamma_0}{\Delta t} \tilde{\mathbf{u}}_m - \hat{\nabla}_m p_m^{n+1}\right). \quad (39)$$

The Neumann boundary condition for the Poisson Eq. (9) becomes:

$$\begin{aligned} \left(\frac{\partial p^{n+1}}{\partial n}\right)_m &= \mathbf{n} \cdot \left(\frac{\partial \mathbf{u}^{n+1}}{\partial t}\right)_m - \beta_0 \mathbf{n} \cdot \left(FFT_m\{((\mathbf{u}^n - \mathbf{w}^n) \cdot \nabla) \mathbf{u}^n\} + \frac{1}{\text{Re}} \nabla \times \omega_m^n\right) \\ &\quad - \beta_1 \mathbf{n} \cdot \left(FFT_m\{((\mathbf{u}^{n-1} - \mathbf{w}^{n-1}) \cdot \nabla) \mathbf{u}^{n-1}\} + \frac{1}{\text{Re}} \nabla \times \omega_m^{n-1}\right) \end{aligned}$$

with $\omega_m = \left(\frac{\partial w}{\partial y} - i\beta m v, i\beta m u - \frac{\partial w}{\partial x}, \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$ with $\mathbf{u} = (u, v, w)^T$. Throughout this work, only homogeneous geometries in the Fourier direction (i.e. the z -direction) have been considered and thus $\mathbf{n} = (n_x, n_y, 0)^T$.

To remove undesired aliasing errors, when computing the non-linear terms, a 3/2 de-aliasing rule [32] is applied in the Fourier direction.

2.4. Implementation, solution of the linear system and parallelisation

The method described has been implemented using the object oriented language C++, which allows for complex-number arithmetics.

External libraries have been used to improve the solver performance. Namely, matrix–vector operations are computed using the intel version of the BLAS libraries and solution of the linear systems resulting from the DG-SIPG discretisation are computed using the direct-iterative intel MKL-Pardiso solver [33] in its sparse version. In addition, to compute the required Fast Fourier Transforms, the FFTW libraries [34] have been used.

To alleviate the increased computational cost associated with computing three dimensional solutions of the full incompressible Navier–Stokes equations, a parallelisation strategy based on the Message Passing Interface (MPI) paradigm is implemented. To parallelise the present solver, the authors followed the parallelisation strategy proposed in [35] for a conformal h/p spectral-Fourier solver. The solution of the incompressible 3D NS equations requires computation of $M/2 + 1$ independent solves Eqs. (38) and (39) and of non-linear terms Eq. (36). A natural parallelisation strategy takes advantage of this decoupling and assigns an equal number of linear solves to each computational node. This strategy ensures that the only necessary communication is performed at the beginning and end of each step and before computation of the non-linear terms. In addition, one can obtain further performance improvement if the computation of the non-linear terms is parallelised. This is achieved, in the present implementation, through a domain decomposition strategy, where each computational node computes the non-linear terms in physical space over an equal number of mesh elements.

3. Two dimensional flows

This section is devoted to code verification and determining the convergence properties of the described method in two dimensions (i.e. x – y DG plane). We first select the Taylor vortex problem [1], where the analytical solution is known to explore the numerical characteristics of triangular–quadrilateral element mixing, internal curved edges and sliding mesh interfaces. Secondly, we compute solutions on a static (non-rotating) NACA0015 at zero angle of attack to explore the solver convergence properties with relation to the NACA 4-digit boundary mapping. In addition, we explore the solver's ability to compute a rotating NACA0015 airfoil and 2D cross-flow turbines under unconstrained and ducted conditions.

3.1. Taylor vortex problem

The first selected problem is the so-called Taylor vortex [1] which represents the evolution of time decaying standing vortices. The time-dependent solution that is used to impose Dirichlet boundary conditions and the initial condition is given by:

$$\begin{aligned} (u, v) &= (-\cos(\pi x) \sin(\pi y), \sin(\pi x) \cos(\pi y)) e^{(-2/\text{Re})\pi^2 t}, \\ p &= -\frac{1}{4}(\cos(2\pi x) + \cos(2\pi y)) e^{(-4/\text{Re})\pi^2 t}. \end{aligned} \quad (40)$$

The problem is solved in $\Omega = [-1, 1]^2$ using various meshes. We select the Reynolds number $\text{Re} = 50$ and run all simulations up to a non-dimensional time $t = 0.1$.

The solver's accuracy is quantified in term of the L^2 relative error norm, which is defined for velocity and pressure as:

$$\frac{\|\mathbf{u}^{\text{exact}} - \mathbf{u}_h^{\text{DG}}\|_{L^2(\Omega)}}{\|\mathbf{u}^{\text{exact}}\|_{L^2(\Omega)}}, \quad \frac{\|p^{\text{exact}} - p_h^{\text{DG}}\|_{L^2(\Omega)}}{\|p^{\text{exact}}\|_{L^2(\Omega)}}. \quad (41)$$

3.1.1. Non-rotating cases

The first verification cases involve only non-rotating meshes (i.e. Eulerian approach) to assess the solver accuracy on mixed triangular–quadrilateral element meshes with internal curved edges and hanging nodes.

We fix $\Delta t = 0.001$ and depict in Fig. 4 the meshes and the L^2 relative error norm convergence for p -refinement obtained using six different static meshes:

- (a) a semi-structured triangular element mesh with 72 elements,
- (b) a structured quadrilateral element mesh with 36 elements,
- (c) an unstructured triangular element mesh with 86 elements,
- (d) an unstructured triangular–quadrilateral element mesh with 100 elements,
- (e) an unstructured triangular–quadrilateral element mesh with internal curved edges and 100 elements,
- (f) an unstructured triangular–quadrilateral element mesh with internal curved edges and hanging nodes with 100 elements.

We observe exponential convergence for pressure and velocity in all cases.

Fig. 4(d) shows that the *functional* incompatibility arising from the use of triangular–quadrilateral elements does not damage the exponential convergence. Similarly, in Fig. 4(e), the methodology presented for the imposition of internal curved edges maintains the high order properties of the method. Finally, in Fig. 4(f), we show that the *geometric* incompatibility arising from the existence of hanging nodes can be handled by the solver without damaging the exponential decay of the error.

3.1.2. Rotating cases

The mesh depicted in Fig. 4(e) and 4(f) is now used together with the ALE formulation and the sliding mesh capability.

We first explore p -refinement, set a small value for the time step $\Delta t = 0.0005$ and run the simulations for a range of rotational speeds $\omega U = 0, 1, 5, 10$ and 15 . We show the results for polynomial enrichment in Fig. 5, where we observe exponential convergence in the L^2 norm for all cases for $k \leq 5$. However it appears that as the rotational speed is increased, the L^2 relative error for high polynomial orders increases. This is found to be related to the temporal error as explained below.

We now fix the polynomial order $k = 7$ and perform a time step refinement study. Results for the L^2 relative error norm for pressure and velocity and for various rotational speeds are plotted in Fig. 6. The slopes in the figure are calculated using the four points corresponding to higher time steps and show second order $\mathcal{O}(\Delta t^2)$ temporal accuracy for pressure and velocity for all rotational speeds.

Theoretical temporal convergence rates, for this splitting scheme, in the L^2 norm for a Stokes problem were obtained in [36]: $\mathcal{O}(\Delta t^{3/2})$ for pressure and $\mathcal{O}(\Delta t^2)$ for velocity, which are in good agreement for a Stokes problem studied by the authors [10]. However, in this test case for the Navier–Stokes equations we see an improvement on the convergence rate for the pressure; $\mathcal{O}(\Delta t^2)$ are obtained for pressure and velocity for all rotational speeds. More precisely, we observe that $L^2(\Delta t) = \mathcal{O}((1 + \omega^2)\Delta t^2)$, showing that to reach a certain level of accuracy it is necessary to reduce the time step, if a high rotational speed is to be computed.

Returning to the exponential convergence limit for p -refinement shown in Fig. 5, it appears clear from consulting Fig. 6 that the limit in exponential convergence for polynomial enrichment for $\omega > 0$ corresponds to the temporal error when this dominates over the spatial error.

To close this section and for completeness, we analyse the L^2 relative error norm for pressure and velocity plotted against the mesh angle: $\Delta t \omega / U$ in Fig. 7 which shows the second order accuracy of the scheme, consistently with the observations of Fig. 6 and confirms the L^2 error estimate: $L^2(\Delta t) = \mathcal{O}((1 + \omega^2)\Delta t^2)$.

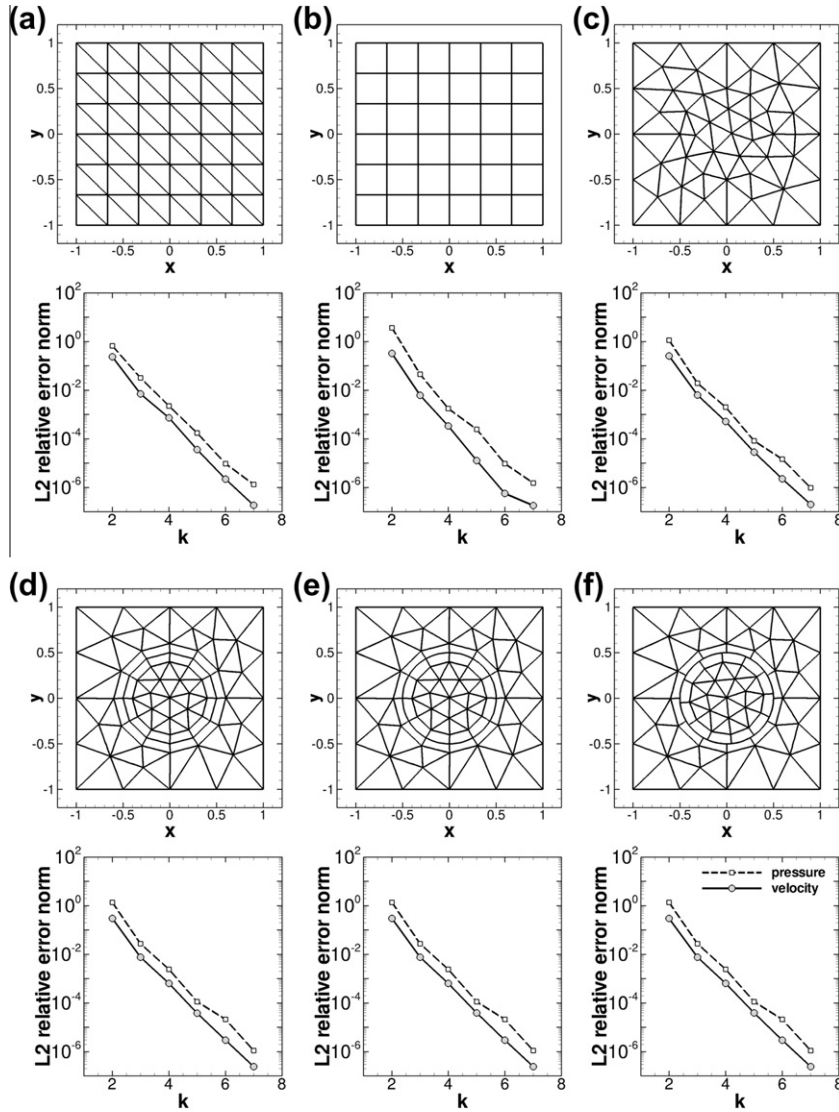


Fig. 4. L^2 relative error norm of pressure and velocity for the Taylor-vortex problem for p -refinement (polynomial enrichment) in various meshes; k represents the polynomial order.

3.2. NACA0015 airfoil

The symmetric NACA0015 airfoil section is selected for verification. Firstly, static cases are considered to assess the implementation and accuracy of the NACA 4-digit boundary condition (defined in Section 2.2.4). Secondly, we consider the NACA0015 under rotation, where we combine the sliding mesh capability and the NACA4-digit boundary condition.

3.2.1. Non-rotating cases

In this section we test the implementation of the NACA 4-digit boundary condition Eq. (21) and check the solver convergence properties using a NACA0015 at zero angle of attack (AOA) and two Reynolds numbers (i.e. $Re = 100$ and 500). For all cases we ensure that the inflow boundary is at 6 chords upstream of the airfoil and the outflow boundary is at 13 chords aft the airfoil. On the upper and lower surfaces, we set Dirichlet boundaries for velocity, and ensure that they are at 8 chords from the airfoil.

We require a definition for the pressure and viscous forces on the airfoil in terms of Cartesian coordinates [1,24], which we define as: $\mathbf{F} = \mathbf{F}_p + \mathbf{F}_v = (F^x, F^y)^T = (F_p^x + F_v^x, F_p^y + F_v^y)^T$ with :

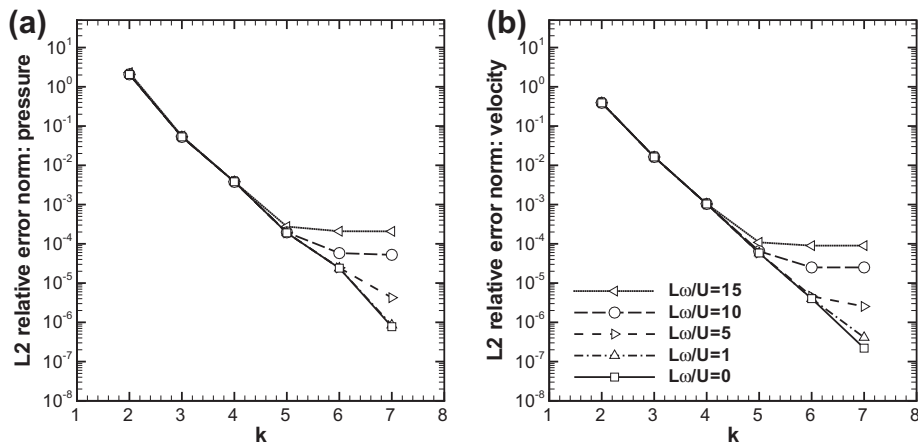


Fig. 5. L^2 relative error norm for p -refinement, Taylor vortex problem with sliding meshes and various rotational speeds for (a) pressure and (b) velocity; k represents the polynomial order.

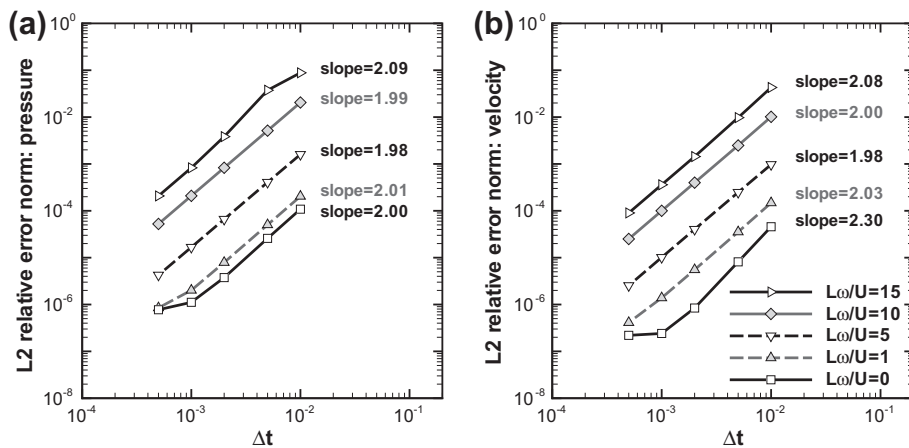


Fig. 6. L^2 relative error norm for time step refinement, Taylor-vortex problem with sliding meshes and various rotational speeds for (a) pressure and (b) velocity.

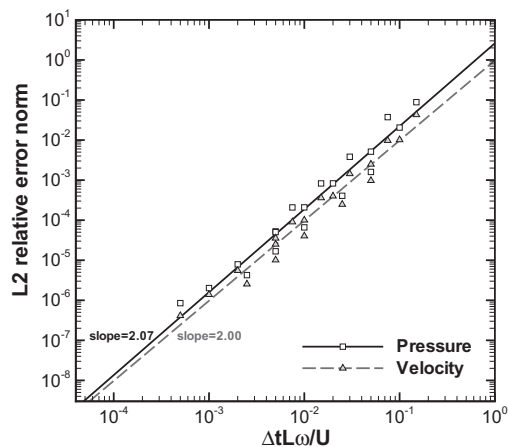


Fig. 7. L^2 relative error norm of pressure and velocity for non-dimensional mesh angle ($\Delta t L\omega/U$) refinement, Taylor-vortex problem with sliding meshes and various rotational speeds.

$$\begin{aligned}\mathbf{F}_p &= \oint p \mathbf{n} ds \\ \mathbf{F}_v &= - \oint v [\nabla \mathbf{u} + (\nabla \mathbf{u})^T] \cdot \mathbf{n} ds,\end{aligned}\quad (42)$$

where $\mathbf{n}$ is the outward pointing normal at airfoil walls.

Firstly, we compare two simulations using the 2D DG solver for a mesh constituted of straight edged triangles and a triangular–quadrilateral mesh that uses the analytical NACA boundaries, Eq. (21). The triangular mesh has a total of 988 elements of which 64 are fitted to the airfoil surface to define the geometry. The triangular–quadrilateral mesh with curved NACA boundaries requires a total of 825 elements and 48 curved quadrilateral elements to define the airfoil geometry. The latter mesh is shown in Fig. 9. We simulate both cases with a polynomial order $k=5$ and depict in Fig. 8, the pressure coefficient ($C_p = 2p$) and the skin friction coefficient ($C_f = 2\tau_w/\text{Re}$ with τ_w the shear stress at the wall) for the NACA0015 at $\text{Re} = 500$ and zero angle of attack ($\text{AOA} = 0^\circ$). Fig. 8(b) shows a smooth distribution along the chord for both variables when the curved parametrisation is used as opposed to the jagged curve obtained with straight edges Fig. 8(a), verifying the necessity of the NACA curved boundaries and its correct implementation. We note that the use of NACA boundaries allows the use of fewer elements to define the airfoil geometry whilst increasing the resolution of aerodynamic quantities.

We then proceed to investigate the solver convergence properties for two Reynolds numbers $\text{Re} = 100$ and 500 at zero angle of attack, $\text{AOA} = 0^\circ$. We use the non-symmetric (with respect to the horizontal axis) triangular–quadrilateral mesh with curved NACA boundaries described previously (825 elements). We depict the mesh and pressure contours for polynomial orders $k=3$ and 6 in Fig. 9 and see qualitatively good agreement between the two polynomial orders and the expected smoother solution for $k=6$.

Since the lift force for a symmetric airfoil at $\text{AOA} = 0^\circ$ is identically zero, we can perform a p -refinement study to assess the solver convergence properties in a realistic case. We show in Fig. 10 that exponential convergence is obtained for the absolute values of the pressure (F_p^y) and viscous (F_v^y) component of the lift force for the two tested Reynolds numbers.

3.2.2. Rotating cases

In this section we consider cases where the NACA 4-digit boundary is used conjointly with the sliding mesh technique (mesh depicted in Fig. 1 and solution snapshots in Fig. 11). A NACA0015 airfoil under impulsive start conditions is rotated around its quarter chord point for a range of geometric angles of attack (i.e. the angle between the free stream velocity and the airfoil chord line) $\text{AOA} = 0^\circ$ – 35° and four rotational speeds $L\omega/U = 0.005, 0.01, 0.05$ and 0.1 , where $L = c$ and c represents the airfoil chord. The Reynolds number is fixed to $\text{Re} = 100$. We use a triangular–quadrilateral mesh constituted by 1462 elements and fix the polynomial order to $k=5$. We show in Fig. 11 snapshots of the unsteady simulation for $L\omega/U = 0.05$ at two geometric AOAs ($\text{AOA} = 17.2^\circ$ and 28.6°). We observe that the contours are smooth across the interfaces between triangular and quadrilateral elements and also across the sliding mesh interface.

Finally, we plot the variation of the vertical F^y and horizontal F^x forces (including pressure and viscous components in Eq. (42)) against the geometric AOA for the four rotational speeds considered in Fig. 12. In addition, the time averaged forces for the static cases for $\text{AOA} = 0^\circ, 10^\circ, 20^\circ$ and 30° have been included. Comparison between static cases and dynamic simulations at low rotational speed show very good agreement demonstrating the low error level introduced by the rotating mesh

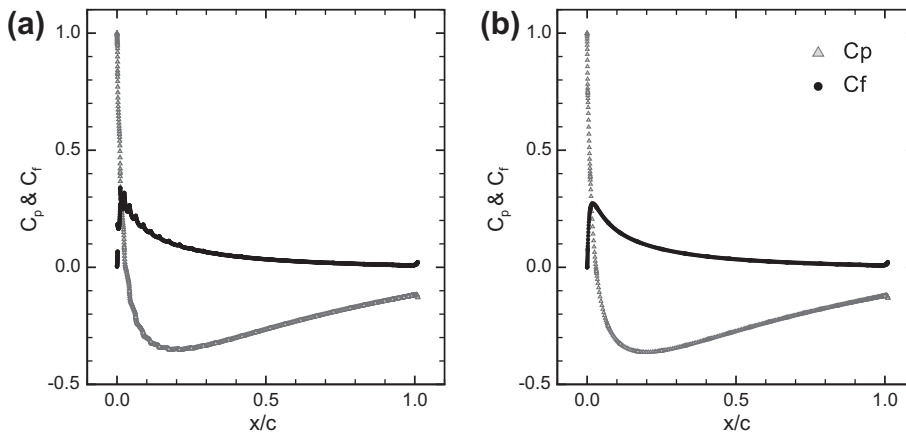


Fig. 8. Pressure (C_p) and skin friction (C_f) coefficients for the NACA0015 airfoil at $\text{Re} = 500$, $\text{AOA} = 0^\circ$ and polynomial order $k = 5$ for (a) Triangular elements with straight edges and (b) Quadrilateral elements with NACA 4-digit curved boundaries.

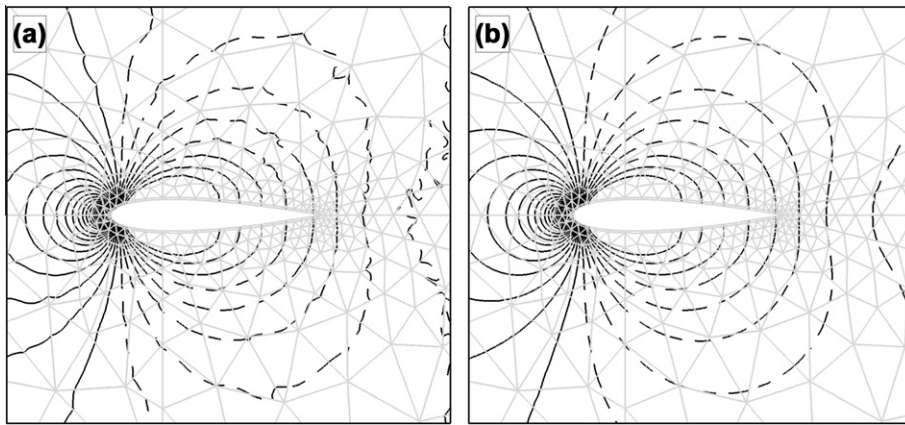


Fig. 9. Solution snapshots of a NACA0015 airfoil at $Re = 500$ and $AOA = 0^\circ$ with 35 pressure contours $[-0.15:0.5]$ (dashed lines show negative values) for polynomial orders (a) $k = 3$ and (b) $k = 6$.

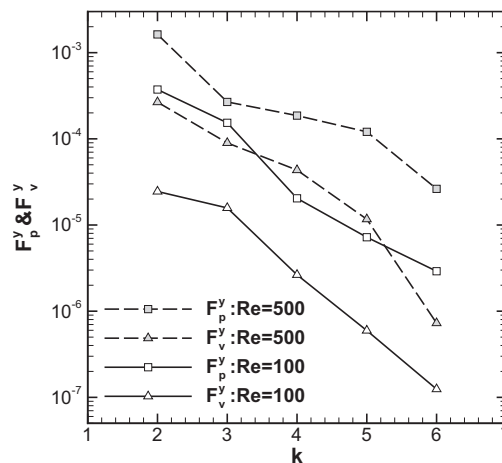


Fig. 10. Exponential convergence for the absolute value of the pressure and viscous components of the lift force (F_p^y and F_v^y) for the NACA0015 at $Re = 100$ and 500 , and $AOA = 0^\circ$; k represents the polynomial order.

interface. We also note that dynamic effects cause an increase in the vertical and horizontal components of the forces with increasing rotational speed.

3.3. An application: three bladed cross-flow turbines

This section presents the solver's ability to compute flows through cross-flow turbines (Darrieus type) in two dimensions. This type of turbine exhibits azimuthal changes in blade aerodynamics, leading to complex flow-phenomena such as stalled flows, vortex shedding and blade-vortex interaction.

We consider two configurations: an unconstrained three bladed turbine and a three bladed turbine placed in an elliptically shaped duct (constrained or ducted case). In addition to the classic unconstrained configuration, the ducted case has been selected to demonstrate the ability of the sliding mesh technique to simulate the relative rotational motion of the inner mesh (rotating with the turbine blades) with respect to the outer static mesh (stationary with respect to the duct walls).

The three bladed turbine configuration consists of NACA0015 airfoils located at a radial distance from the center of rotation of two chords $R = 2c$. The relative angle between blades is fixed at 120° as shown in Fig. 13. The rotational speed is set to $L\omega/U = 0.5$, where U is the free stream velocity magnitude and $L = c$. The Reynolds number based on the free stream velocity and the blade chord is $Re = 50$. We note that at this low Reynolds number, three dimensionalities may be disregarded, enabling purely 2D flow simulations.

The polynomial order is fixed to $k = 3$, the time step Δt to 0.0005 and we use two triangular–quadrilateral meshes with curved NACA boundaries constituted of: 3320 elements for the unconstrained three bladed turbine and 3614 for the ducted three bladed turbine. We depict in Fig. 13(a) and (b), snapshots of the velocity magnitude for the unconstrained and ducted

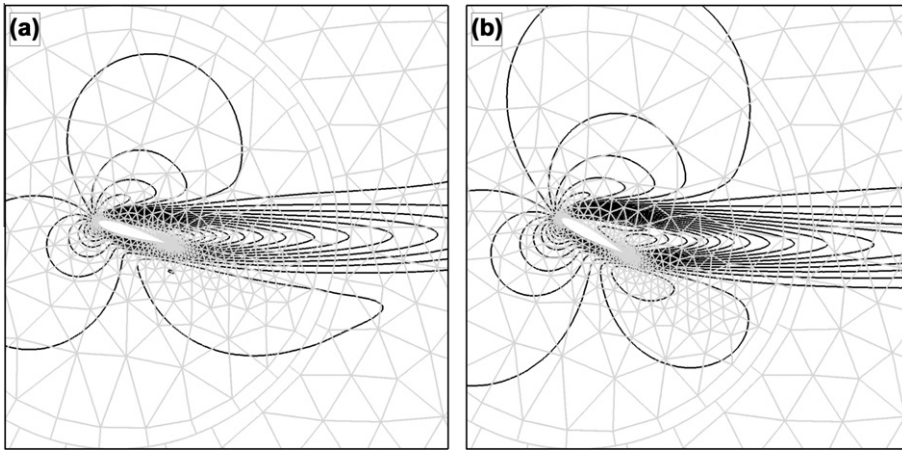


Fig. 11. Solution snapshots of the rotating NACA0015 with 21 velocity magnitude contours [0:1.3] for polynomial order $k = 5$ and rotational speed $L\omega/U = 0.05$ for (a) geometric $AOA = 17.2^\circ$ and (b) geometric $AOA = 28.6^\circ$.

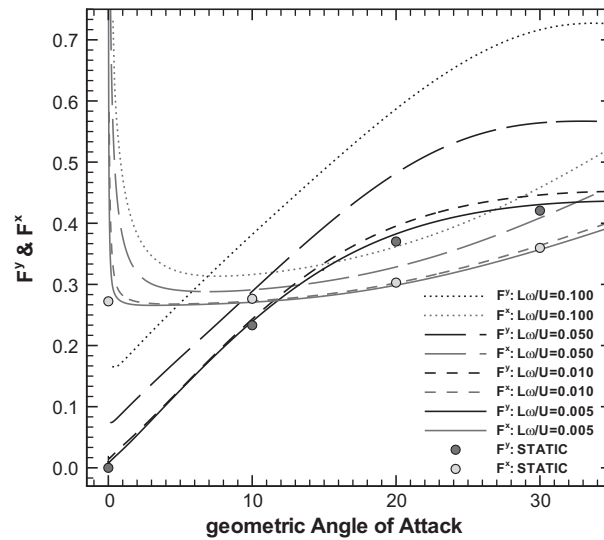


Fig. 12. Vertical and horizontal forces (F^y and F^x) against the geometric $AOA = 0^\circ$ to 35° , for the NACA0015 airfoil at various rotational speeds $L\omega/U = 0.005, 0.01, 0.05$ and 0.1 . Time averaged forces for static simulations are included for $AOA = 0^\circ, 10^\circ, 20^\circ$ and 30° .

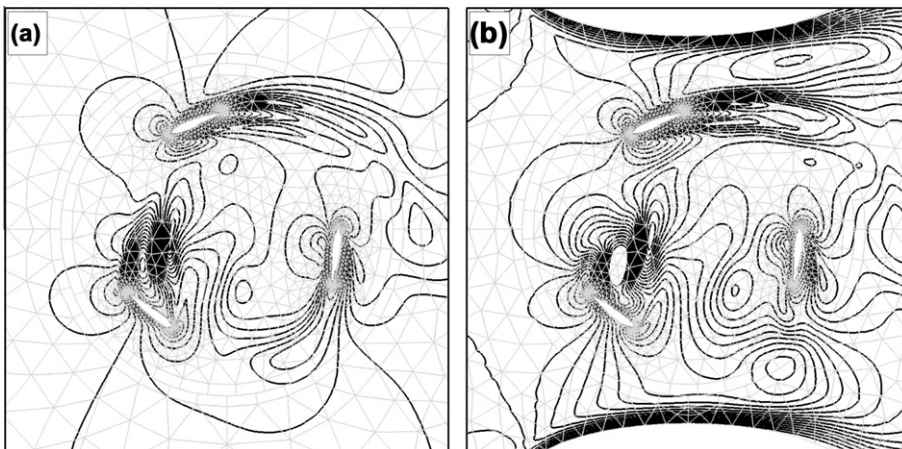


Fig. 13. Solution snapshots of a three-bladed cross-flow turbine with 21 velocity magnitude contours [0:1.5] for polynomial order $k = 3$ for (a) unconstrained and (b) ducted cases.

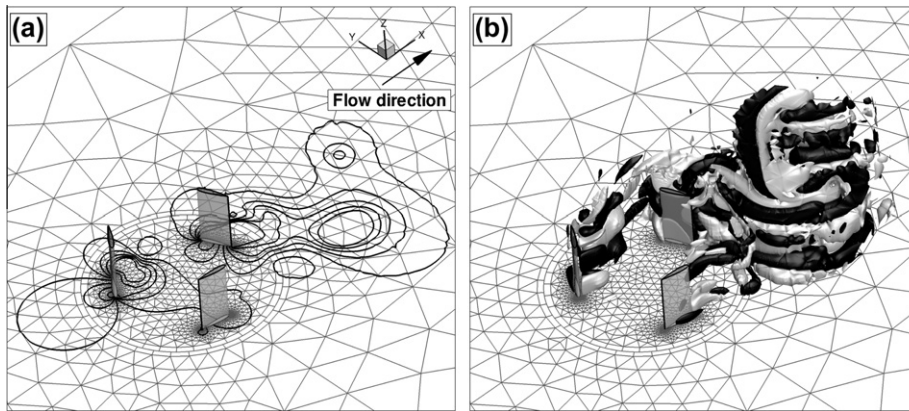


Fig. 14. Solution snapshots of flow through a 3D cross-flow turbine: (a) 11 pressure contours $[-1.6:0.8]$ and (b) iso-surfaces of z -velocity $w = \pm 1 \times 10^{-6}$.

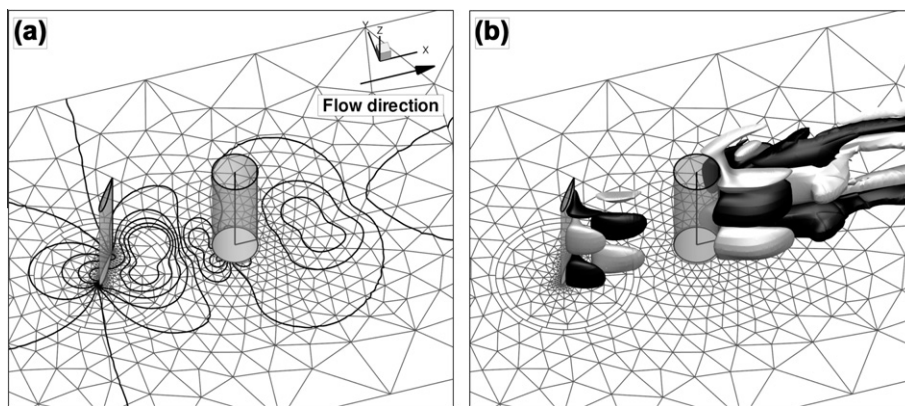


Fig. 15. Solution snapshots of a 3D circular cylinder shadowed by a rotating NACA blade (a) 11 pressure contours $[-0.6:0.6]$ and (b) iso-surfaces of z -velocity $w = \pm 1.5 \times 10^{-6}$.

devices respectively. It can be seen that the DG solver is able to resolve the complex flow physics arising from the rotational motion, loading and unloading of the blades, and the interaction with the boundary layers of the inertially fixed duct walls.

4. Three dimensional flows

In this last section, we provide two examples of three dimensional flow simulations using the Fourier extension of the DG solver with rotating sliding meshes.

Firstly, the three bladed cross-flow turbine geometry described in Section 3.3 is simulated with a rotational speed of $L\omega/U = 0.1$ at a Reynolds number (based on the freestream velocity and blade chord) of $Re = 200$ (i.e. large enough to trigger three dimensionalities in the flow). The spatial discretisation consists of polynomials of order $k = 4$ in the x - y DG plane and 16 Fourier planes in the z -direction to discretise a spanwise length of $L_z/c = 2$, where c is the blade chord. Fig. 14(a) shows pressure contours and Fig. 14(b) depicts iso-surfaces of z -velocity showing the three dimensional character of the solution.

Secondly, we present a problem where a circular cylinder is shadowed by a pitching NACA0012 airfoil. This test case is presented to illustrate the possibility of locating static and rotating 3D geometries together in a single mesh domain. In addition, all of the features presented in this paper are included in a single simulation: high order sliding meshes and external curved boundary condition for NACA airfoils and circular cylinders.

The NACA airfoil is pitching with a rotational speed of $L\omega/U = 0.5$ and the Reynolds number based on the airfoil chord (which is equal to the cylinder diameter) is $Re = 200$. As in the previous 3D case, the spatial discretisation consists of polynomials of order $k = 4$ in the x - y DG plane and 16 Fourier planes in the z -direction. The spanwise length is, in this case, set to $L_z/c = 2.5$. Fig. 15 shows the pressure contours and iso-surfaces of z -velocity. It can be seen that the sliding mesh technique does not damage the numerical solution and captures the pressure interactions between static and rotating geometries. In addition, Fig. 15(b) shows that three dimensionalities are present in the wake of both geometries.

5. Conclusions

This work details the formulation, implementation and verification of a novel approach to solve the three dimensional incompressible NS equations using a high order DG-Fourier formulation with rotating sliding meshes. The solver allows accurate solutions with curved elements that conform to either arced surfaces (internal or external) and the profile of NACA 4-digit airfoils. Hybrid unstructured meshes (i.e. with triangular and quadrilateral elements) can be handled without damaging the solver's high order numerical properties. To account for relative motion, the NS equations are written in Arbitrary Lagrangian–Eulerian form and the sliding interface is given special treatment that maintains the temporal convergence rates and spectral properties of the code when rotational motion is applied to a region of the mesh. In addition, it has been shown that the high order properties are maintained when using body fitted NACA 4-digit boundaries.

We have demonstrated that *geometric* and *functional* incompatibilities do not represent a problem in the DG context and that sliding mesh interfaces can be handled naturally without the necessity of *mortar* type techniques.

The solver accuracy has been verified with three test cases for two dimensional flows: the Taylor vortex problem, a static and rotating NACA0015 airfoil and a three bladed cross flow turbine under unconstrained and ducted flow conditions. These test cases have enabled verification of the code as well as demonstrating its capabilities to simulate flow problems in which accuracy is mandatory.

Three dimensional flow features are captured through a Fourier extension that allows high order 3D solutions for rotating geometries that present a spanwise geometric homogeneity. 3D flow simulation examples are provided for a three bladed cross-flow turbine and a pitching airfoil shadowing a circular cylinder.

The presented two and three dimensional simulations have demonstrated the solver's capability to compute complex flow physics for problems where static and rotating geometries coexist in a single domain. We conclude that the novel approach to account for sliding mesh interfaces and curved external boundaries provides a promising framework for studying fluid–structure interaction problems involving rotation.

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